STUDENT RECORD CLI MANAGER

Final Project Submission

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PROBLEMS

Managing student records manually can be difficult and inefficient.

There is a need for a simple system to:

- Store student data
- Retrieve information easily
- Analyze student performance

PROPOSED SOLUTION

A command-line Student Management System that:

- Allows adding student records
- Stores data in a JSON file
- Provides search and delete functionality
- Calculates average score and top performer

KEY FEATURES

- Add Student
- View Students
- Search Student
- Delete Student
- Analyze Data (Average & Top Student)

DATA HANDLING

- Data stored in JSON file (students.json)
- Each student has:
  - Name
  - Score
- Data is loaded and saved dynamically

Program Output
(Menu & Result)

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Khalifaxx\Documents\Python Programs\student-management-system> py main.py

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View Students
3. Search Student
4. Delete Student
5. Analyze Data
6. Exit
Choose option:
```

```
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View Students
3. Search Student
4. Delete Student
5. Analyze Data
6. Exit
Choose option: 1
Enter student name: Ahmad Ahidjo
Enter score: 76
✅ Student added!

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View Students
3. Search Student
4. Delete Student
5. Analyze Data
6. Exit
Choose option: 2
1. Usman  - 65.0
2. Usman Mustapha - 65.0
3. Ibrahim Muhammed - 49.0
4. Ahmad Ahidjo - 76.0

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View Students
3. Search Student
4. Delete Student
5. Analyze Data
6. Exit
Choose option: 5
📊 Average Score: 63.75
🏆 Top Student: Ahmad Ahidjo (76.0)
```

CONCLUSION

- The system successfully manages student data
- It simplifies record keeping and analysis
- Demonstrates use of Python concepts such as:
  - Functions
  - File handling
  - Data processing

Thank You!

Questions?